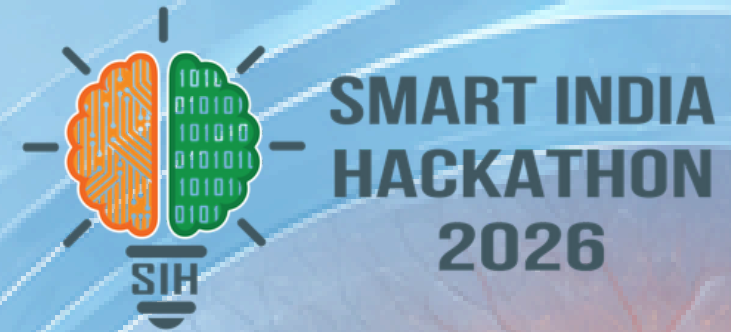


SMART INDIA HACKATHON 2026



PROBLEM STATEMENT ID – SIH26038

PROBLEM STATEMENT - EXPLAINABLE AI FOR DIABETIC RETINOPATHY

SCREENING IN RURAL INDIA

THEME- MEDTECH / BIOTECH / HEALTHTECH

PS CATEGORY- SOFTWARE

TEAM ID- 154299

TEAM NAME (REGISTERED ON PORTAL): SIXTH_SENSE_CODERS

[DEMO LINK](#)

[VIDEO LINK](#)

[DOCTOR VALIDATION](#)

[CAD FUNDUS DESIGN](#)

[MOBILE/ UI INTERFACE](#)



PROBLEM STATEMENT



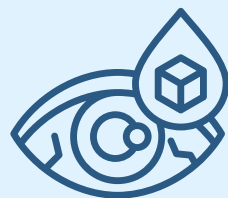
THE CONTEXT

Diabetic Retinopathy Screening in Rural India

Millions of people in rural and resource-constrained areas are at risk of vision loss due to late detection of Diabetic Retinopathy (DR).



India has
~100
Million+
people with
diabetes.



~1 in 3
diabetic
patients
develop DR.



Many cases
are detect-
ed at a late
stage.



KEY CHALLENGES



Limited Access to Specialists

Rural areas have few ophthalmologists and screening facilities



Poor Image Quality

Many fundus images are blurred or incomplete, affecting reliable screening.



Connectivity Challenges

Unstable internet limits the use of cloud-based screening solutions.



Delayed Intervention

Many cases are detected late, leading to preventable vision loss.

RETINASCAN AI: OUR SOLUTION



Quality Aware Screening

Automatically checks image quality and guides re-capture when needed.



AI Assisted Analysis

Detects DR severity and highlights possible lesions (microaneurysms, haemorrhages, exudates, etc.).



Explainable Results

Shows visual evidence (heatmaps / lesion highlights) to support clinical understanding.



Doctor in the Loop

Ophthalmologists can confirm, modify or reject the AI result.



Patient Friendly Output

Generates structured reports and supports multilingual guidance for better accessibility.



OUR GOAL

Make early DR screening simple, reliable and accessible for every community, everywhere.



Detect
Earlier



Enable
Trust



Reach More
People



Prevent
Vision Loss

[DEMO LINK](#)



1. USER ACCESS & WEB APPLICATION LAYER(For All Users)

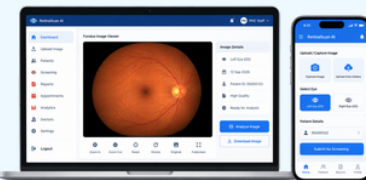
USERS: RURAL HEALTHCARE SETUP



Patient / Screening Staff

- Capture / Upload Fundus Image
- Left / Right Eye
- Basic Details
- Submit for Screening

WEB APPLICATION: REACT PWA



- Responsive — Mobile / PC
- Easy to Use Interface
- Upload / Capture Fundus Image
- View Screening Results
- Patient / Doctor / Admin Access

USER ROLES

- Screening User: Patient / PHC Staff
- Upload / Capture Image
 - View Screening Result
 - Download report

- Doctor — Ophthalmologist
- View & Verify Cases
 - See Grad-CAM and Lesions
 - Modify / Confirm Results
 - Manage Patient Records

- Admin / Program Manager
- Screening Statistics
 - Patient Volume
 - Resource Utilization
 - Manage Centres and Users

WEB UI SCREENS (EXAMPLES)



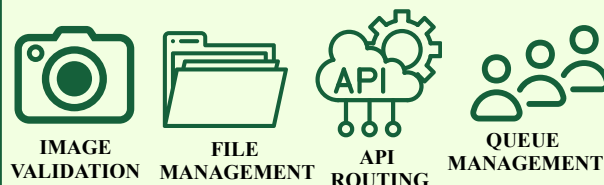
2.BACKEND & MATLAB INTEGRATION LAYER

WEB BACKEND: FASTAPI

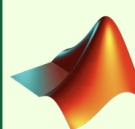


- Handle Image Requests
- Manage User & Patient Records
- Call MATLAB Analysis Pipeline
- Store Results & Reports
- Provide APIs to Web UI

REQUEST HANDLER



MATLAB ANALYSIS ENGINE (Core DR Pipeline)



- Image Processing
- Deep Learning
- Computer Vision
- Medical Image Processing Toolbox

REQUEST HANDLER

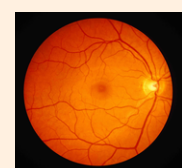


DATABASE & STORAGE

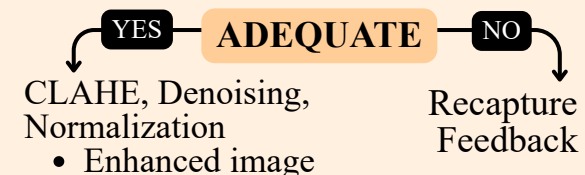
- Patient Records
- Images & Results
- User Data
- Report Storage
- Analytics Data

3. MATLAB RETINAL IMAGE ANALYSIS PIPELINE

IMAGE QUALITY ASSESSMENT & ENHANCEMENT



- Focus / Blur Check
- Illumination Check
- Field of View Check
- Image Quality Score



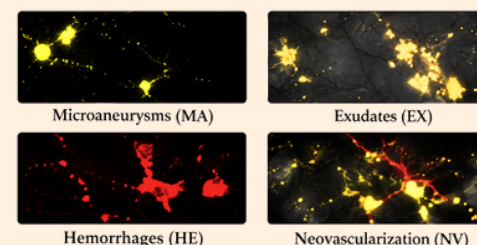
RETINAL STRUCTURE SEGMENTATION



Optic Disc & Fovea Vessel Segmentation

- Optic Disc Localization
- Fovea Localization
- Vessel Segmentation

IMAGE QUALITY ASSESSMENT & ENHANCEMENT



- Microaneurysms (MA)
- Exudates (EX)
- Hemorrhages (HE)
- Neovascularization (NV)

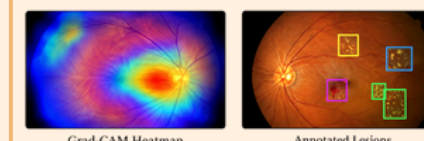
DR SEVERITY GRADING



- 0 · No DR
- 1 · Mild NPDR
- 2 · Moderate NPDR
- 3 · Severe NPDR
- 4 · Proliferative DR

- Deep Learning Model (e.g. ResNet / EfficientNet)
- Trained and Validated on IDRid / APTOS / DRIVE
- Referral if DR $\geq$ Level 2
- Target Sensitivity: $\geq 90\%$
- Target Specificity: $\geq 85\%$

EXPLAINABILITY & CLINICAL REPORT



- Grad-CAM Heatmap
- Lesion-Level Evidence
- Confidence Scores
- Automated Annotated Report — PDF
- Ophthalmologist Validation

4. SIMULINK TELEMEDICINE WORKFLOW SIMULATION



PATIENT ARRIVAL

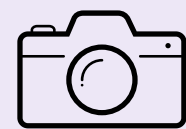
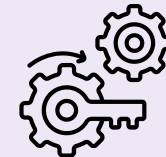


IMAGE ACQUISITION



NETWORK BANDWIDTH



PROCESSING QUEUE



OPHTHALMOLOGIST REVIEW

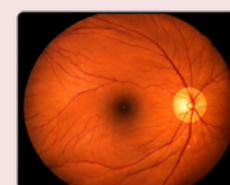


REFERRAL / DECISION

5. DATASETS & VALIDATION (MATLAB)



IDRID (Indian Dataset)



APTOS 2019 (Validation)



DRIVE (Vessel Segmentation)



Messidor-2 (External Validation)

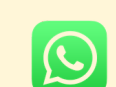
6. OUTPUTS & INTEGRATIONS



Clinical Report (PDF)



Referral & Follow-up (Primary Healthcare)



WhatsApp / SMS Notifications



ABHA / FHIR Integration

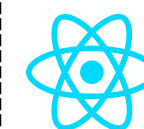


Patient Records (Database)



Analytics Dashboard (Program Monitoring)

TECHNOLOGY STACK



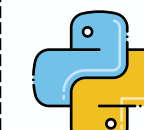
React



Vite



Tailwind CSS



Python



FastAPI



ONNX



OpenCV



MATLAB



Simulink



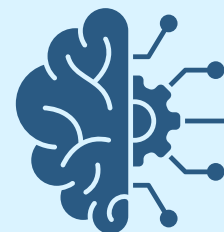
Supabase

FEASIBILITY



MATLAB Based Development

Image processing and deep learning tools support the development of our screening pipeline.



Modular AI Pipeline

Quality checks, lesion detection, severity grading and explainability can be developed and tested as separate modules.



Rural Deployment Potential

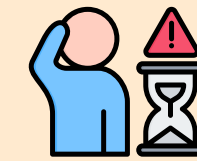
A quality aware workflow can support screening using portable fundus cameras and referral-based care.



Scalable Workflow

Simulink can help study processing time, connectivity and review capacity before wider deployment.

A Path to Real Impact



CONSTRAINTS



Poor Image Quality

Blur, uneven lighting and incomplete retinal views can affect lesion detection.



Small & Difficult Lesions

Microaneurysms and subtle retinal changes may be missed by the model.



Connectivity & Processing Limits

Rural centres may face bandwidth limitations and limited access to computing resources.



STRATEGIES



Quality Aware Screening

Assess image quality, enhance borderline images and request recapture for ungradable images.



Lesion-Focused Analysis

Combine retinal structure segmentation and lesion detection to support severity grading.



Efficient Telemedicine Workflow

Use Simulink to study bandwidth, processing throughput and referral capacity for rural screening.

VIDEO LINK



IMPACT AND BENEFITS



IMPACT

Creating a real difference



FOR PATIENTS

Screen closer to where they live and get timely support for healthier lives.



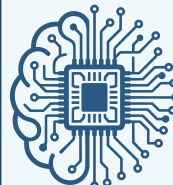
FOR OPHTHALMOLOGISTS

Enables efficient review and better clinical decision making.



FOR PHCS / SCREENING CENTRES

Makes quality screening accessible beyond high end hospitals.



FOR AI RELIABILITY

Builds trustworthy and transparent AI through uncertainty handling.



FOR HEALTH SYSTEMS

Contributes to scalable eye care programs and stronger public health.





BENEFITS

Practical Advantages for all



FOR PATIENTS

- Screening closer to home
- Multilingual, low-connectivity support
- Faster specialist referral



FOR OPHTHALMOLOGISTS

- Pre-screened patient queue
- Image quality flag + AI grade
- Lesion evidence & explainability



FOR PHCS / SCREENING CENTRES

- Offline first screening
- Standardized workflow
- Referral prioritization



FOR AI RELIABILITY

- Poor image → RECAPTURE
- Insufficient evidence → ESCALATE
- AI/lesion disagreement → REVIEW

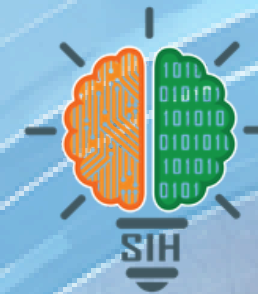


FOR HEALTH SYSTEMS

- Structured screening & referral data
- Audit trail + interoperability pathway
- District-level scalability



RESEARCH AND REFERENCES



1. **PORWAL ET AL. IDRID:**
[LINK](#)
Diabetic Retinopathy Segmentation and Grading Challenge
Supports: DR grading, lesion segmentation, optic-disc/fovea localization, IDRiD validation

2. **MATHWORKS**
[LINK](#)
Multilabel Diabetic Retinopathy Fundus Image Classification Using Deep Learning
Supports: MATLAB, DR classification, preprocessing, CLAHE, denoising, Grad-CAM

3. **GARGEYA & LENG**
[LINK](#)
Automated Identification of Diabetic Retinopathy Using Deep Learning
Supports: automated DR screening and fundus-image classification

4. **QUELLEC ET AL.**
[LINK](#)
Deep Image Mining for Diabetic Retinopathy Screening
Supports: deep-learning based retinal lesion detection and DR screening

5. **SELVARAJU ET AL.**
[LINK](#)
Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization
Supports: Grad-CAM explainability / visual evidence

6. **MATHWORKS**
[LINK](#)
Segmentation of Blood Vessels in Retinal Fundus Images Using Maximum Principal Curvature
Supports: MATLAB retinal vessel segmentation + DRIVE dataset

7. **STAAL ET AL.**
[LINK](#)
Ridge Based Vessel Segmentation in Color Images of the Retina
Supports: retinal blood vessel segmentation and DRIVE benchmark

8. **MATHWORKS**
[LINK](#)
Retinal Blood Vessel Extraction / Image Processing Workflow
Supports: MATLAB image enhancement and retinal vessel extraction

“VALIDATED BY
REAL DOCTORS”

[DEMO LINK](#)
[VIDEO LINK](#)
[DOCTOR VALIDATION](#)
[CAD FUNDUS DESIGN](#)
[MOBILE/ UI INTERFACE](#)

